

CYSTIC FIBROSIS

----- Active space -----

- multi-system **AR** disorder : Affects GIT, RS, Genito-urinary, sweat glands.
- **m/c** life limiting genetic disorder.

Pathogenesis

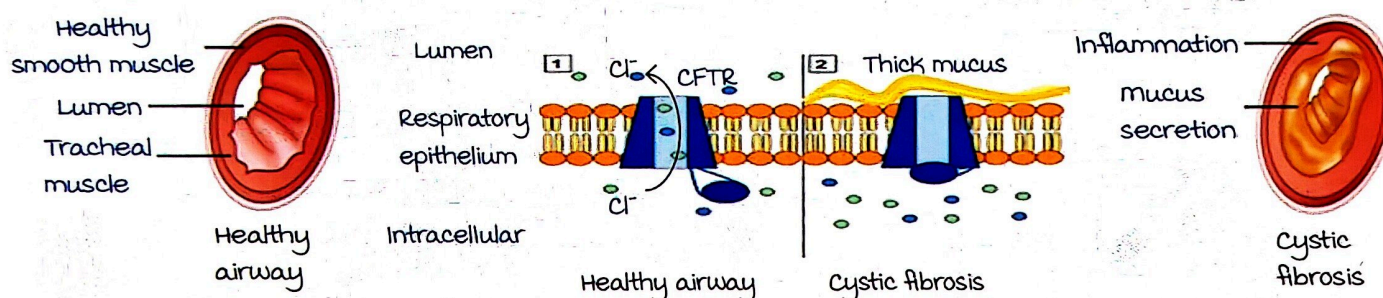
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CFTR gene (CF transmembrane regulator) :

- Location : **Chr 7q**.
- Codes for CFTR protein (Cl^- channel).
- **m/c** mutation : **$\Delta F 508$** → Deletion of phenylalanine at 508th position.

Mucoviscidosis :

Thick mucus secretions in body.



Clinical features

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Gastrointestinal tract features (Earliest) :

I. meconium ileus :

Incidence :

- 10 - 15% patients.
- Virtually diagnostic of CF.

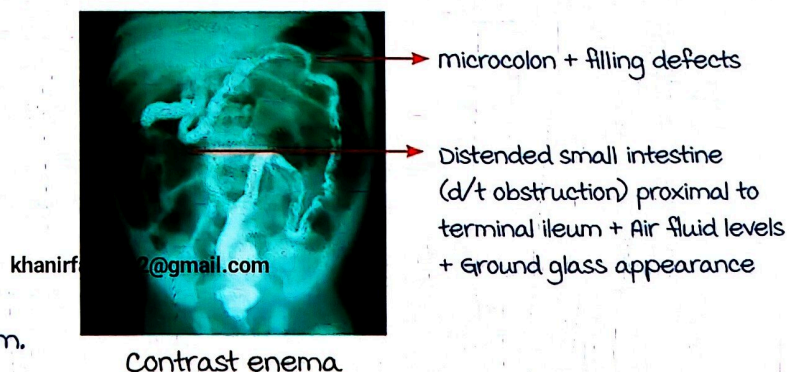
Investigations :

Clinical features :

Inspissation of thick meconium in ileum.

↓
Intestinal obstruction at ileum.

↓
Presents with delayed passage of meconium & abdominal distension.



Contrast enema

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2. Constipation (Older children) : Impaction of fecal matter in distal ileum → **DIOS** (Distal Intestinal Obstruction Syndrome).

3. Pancreatic insufficiency (85% cases) :

Initially exocrine → Later endocrine.

- Exocrine features : Steatorrhea, Vitamin A, D, E & K deficiency.
- Endocrine features (2nd decade) : Diabetes melitus.

4. Rectal prolapse (Older children) : **Rare** d/t early supplementation of pancreatic enzyme.

Respiratory tract features :

1. Recurrent infections by :

- **Staphylococcus aureus** (m/c).
- **H. influenza** type B.

} In early childhood.

• **Pseudomonas** :

- In **late childhood**/adults.
- Causes mucoid secretions → Forms biofilm → Antibiotic resistance.

• **Burkholderia cepacia** : ↑ Risk of **death**.

2. Bilateral nasal polyps.

Other features :

1. Biliary tract : Thick biliary secretions → Bile outflow obstruction → Biliary cirrhosis/neonatal cholestasis.

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2. Genitourinary tract :

- Males : Failure of wolffian duct development → **Azoospermia** → **Infertility**.
- Females : ↓ **Fertility** rate.

3. Sweat glands : Inactive CFTR protein → ↑ loss of Na^+/Cl^- in sweat (↓ Reabsorption) →

- Salty skin (on kissing).
- Frosting of skin.
- Predisposed to **hyponatremic hypochloremic metabolic alkalosis**.
- ↑ in sweat Cl^- test.

Management of cystic fibrosis

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DIAGNOSIS

Clinical criteria : Positive **for any one** → Confirmatory test (Lab test) done.

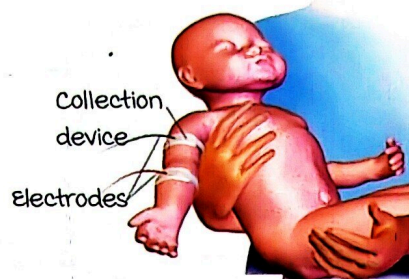
- Typical clinical features : Respiratory/gastrointestinal/genitourinary.
- History of CF in a sibling (AR).
- Positive newborn screening test : Immunoreactive trypsin test (IRT).

Laboratory evidence : Any 1 **positive lab test** confirms CF.

- 2 sweat chloride concentrations (>60 mEq/L) on separate days.
- Identification of 2 CF mutations.
- 1 abnormal value of trans epithelial nasal potential difference.

Pilocarpine Iontophoresis (sweat chloride test) :

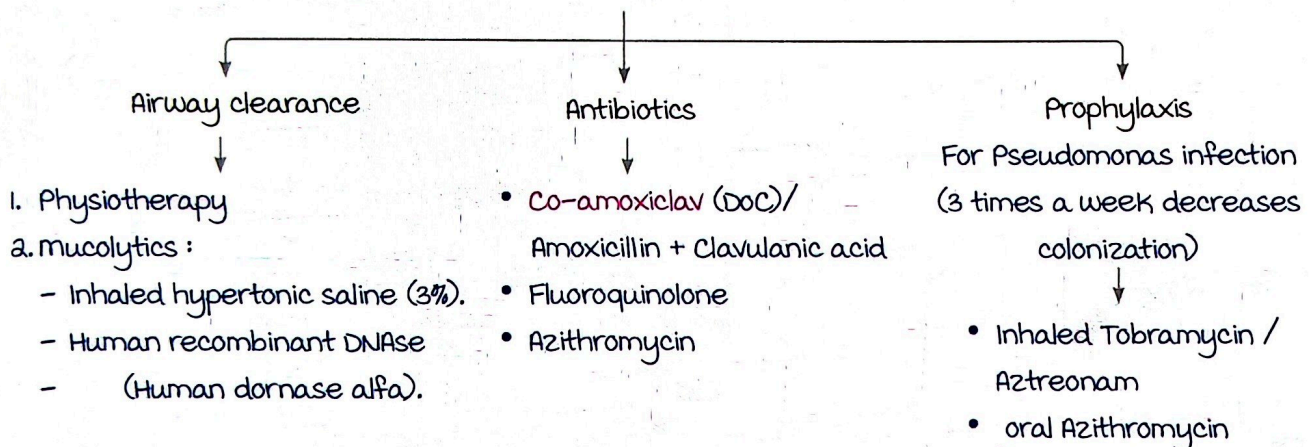
- Pilocarpine administered into skin of baby using electrodes.
- Baby's sweat collected in collection device which processes it for Cl^- levels.



Sweat chloride test

Treatment :

1. Respiratory care :



3. Dietary measures :

- Diet : High fat, protein, calories.
- Salt supplementation (more during dehydration).
- Pancreatic enzyme supplements (enterically coated microspheres).
- Fat soluble vitamin supplementation (A, D, E & K).

4. CFTR modulators ($\uparrow$ activity of CFTR channel) :

- Ivacaftor (>2 years).
- Ivacaftor + Lumacaftor (>6 years) : useful in $\Delta F508$ mutation cases.
- Ivacaftor + Tezacaftor (>12 years).
- Ivacaftor + Tezacaftor + Elexacaftor (**Triple combination therapy**) : Covers most mutations.